

SIH 2024 • SIH1638 • CSE276 • SECTION OJ • GROUP OJ05

AI Crop Disease Prediction & Management

A preliminary screening system that combines leaf images with environmental context.

Hemanth Nidigonda • Deepanshi Sengar • Lovely Professional University

Made with GAMMA



THE SCREENING GAP

Why an early-screening system matters

Similar symptoms

Leaf symptoms look alike; manual inspection takes time.

Changing conditions

Environmental conditions influence disease risk.

Uneven access

Expert access varies across farming contexts.

Right-sized goal

Preliminary awareness and decision support—not a final diagnosis.

Made with GAMMA

TRAINING EVIDENCE

PlantVillage turns leaf images into training evidence



54,306
approximately RGB images

14
crop species

26
diseases

38
classification categories

Image preparation pipeline

Collect → Resize → Normalize → Augment → Split into train / validation / test

Task: multi-class image classification • Dataset scope from PlantVillage and Mohanty et al. (2016)

Made with GAMMA

SYSTEM ARCHITECTURE

Two evidence streams feed one decision surface

Image pathway: Upload leaf → preprocess → CNN / MobileNetV2 → disease classification.

Context pathway: location + time → environmental data → risk assessment.

Final dashboard: prediction, confidence, context, and suggested next steps. Align environmental data by location and time.



Made with GAMMA

MODEL CHOICE

Choose between visual depth and edge efficiency

Compare both after experiments. Do not assume the winner in advance.

CNN

01

Visual depth

Convolution → ReLU → pooling → feature extraction → fully connected layers → softmax.

A deeper feature-learning pathway to test.

MOBILENETV2

02

Edge efficiency

Transfer learning → pretrained features → lightweight computation → efficient inference.

A practical pathway for constrained deployment.

Made with GAMMA

CONTEXT LAYER

Weather adds context, not diagnosis

Air conditions

Temperature • relative humidity

Water conditions

Rainfall • precipitation

Soil conditions

Soil moisture • soil temperature

Possible source

Open-Meteo Historical Weather API

Crop-specific thresholds require agricultural validation.



Made with GAMMA

CREDIBILITY CHECK

Evaluation keeps the prototype credible

01 Split

Training, validation, and held-out test data.

02 Score

Accuracy, precision, recall, F1-score, and confusion matrix.

03 Compare

CNN versus MobileNetV2 on performance, training time, and inference time.

04 Report

[PLACEHOLDER] Show experiment results only after they exist.

No performance values are claimed before the experiments are complete.

Made with GAMMA

SCOPE AND NEXT STEPS

Preliminary awareness, grounded in evidence.

AI, deep learning, and environmental information can support preliminary crop disease identification and risk awareness.

Conclusion, scope, and references

Limitations

Dataset diversity, real-world image quality, alignment, uncertainty, and field validation.

Future

Mobile app, more crops, live weather, multilingual UI, explainability, and expert-verified guidance.

References

PlantVillage Dataset • Mohanty et al. (2016) • MobileNetV2 (CVPR 2018) • Open-Meteo • TensorFlow • scikit-learn

Made with GAMMA